

DSAW Project Description

Problem Statement:

IVF or In Vitro Fertilization is a medical procedure to treat infertility in patients. The dataset has been collected for patients over the years 2020-2021 who have undergone the ICSI procedure. The dataset has 205 rows and 18 columns with various attributes like clinical protocol, medication used for the ICSI procedure with resulting endometrium thickness, no of oocytes retrieved, no of embryos formed and the clinical outcome which is the Beta HCG Result.

Data Dictionary:

IVF No	Patient ID No
Age	Age of the patient
Protocol	Clinical Protocol of medication. 1). Antagonist Type or Short Protocol 2). Lupron Down Regulation 3). Flare Protocol or Micro dose Lupron Co Flare Protocol
Type of patient	Procedure Type
OPU date	Oocyte Pick Up Date
ET Date	Embryo Transfer Date
ET Type	Type of Embryo Transfer 1)Fresh 2) Frozen
GNRH	Type of Gonadotropin Release Hormone name
hmg	Human Menopausal Gonadotropin
RFSH	Recombinant Follicle Stimulating
hcg	Human Chorionic Gonadotropin
endomet	Endometrium Thickness
Eggs Retrieved	No of Eggs retrieved from OPU
Mature Eggs	No of Mature eggs formed
Emb formed	No of Embryos formed
Emb Trans	No of Embryos transferred
Freeze	No of embryos frozen
Beta HCG Result	Pregnancy Result ---Positive/ Negative

Step 1. Read the dataset into the Jupyter notebook using pandas read function

Step 2: Check for shape, datatypes, duplicates, missing values in the dataset. It has missing values in the column "endometrium" and 1 duplicate row. The duplicate row is dropped and the missing values are imputed with median values.

Step 3: Use describe function to get the statistical comparison of the attributes. The mean age of the patients is 32 years. Minimum age is 20 and maximum age is 48 years. The mean endometrium thickness is 9 and the eggs retrieved is 10 with 5 embryos formed.

Step 4: Make boxplot and check for outliers"

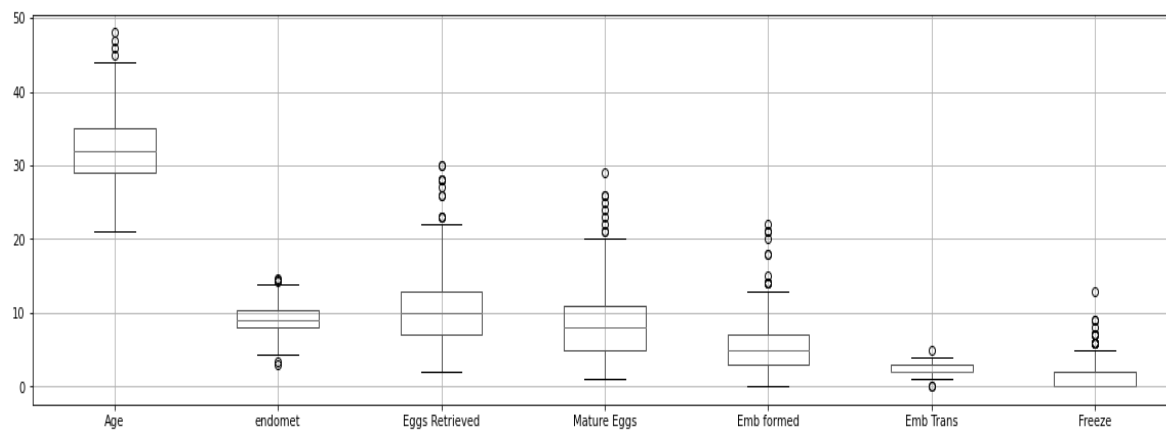


Image 1. Boxplot showing outliers for most of the continuous variables

Step 5: Define function for upper range and lower range removal of outliers.

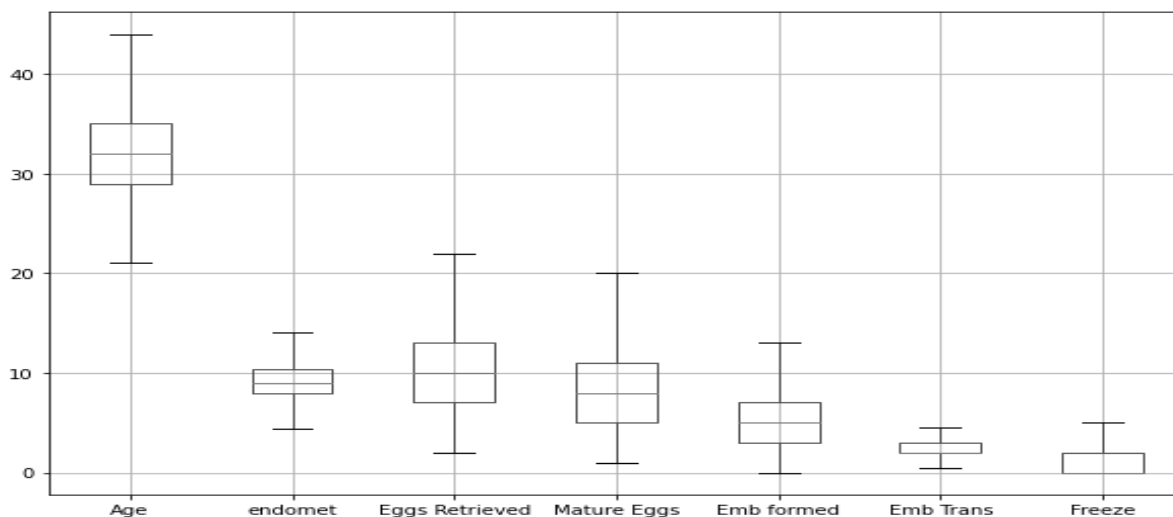


Image 2. Boxplot after removal of outliers

Step 6: Do univariate analysis using distplot for 'Embryos formed', 'Eggs Retrieved' and 'Freeze' attributes.

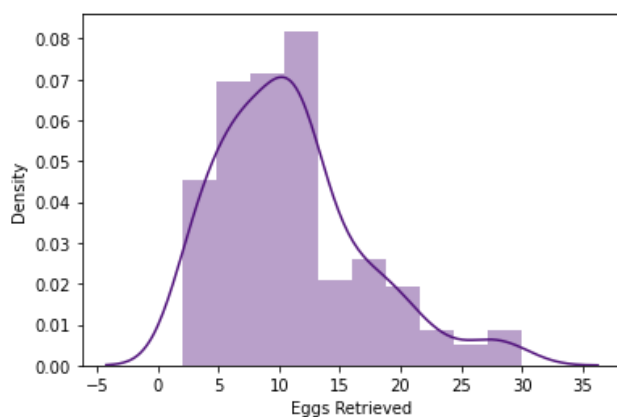


Image 3: Right Skew

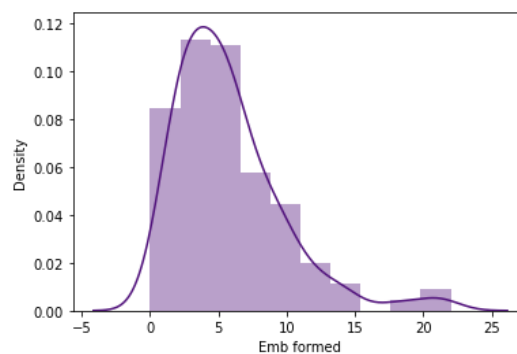


Image 4: Normal Distribution

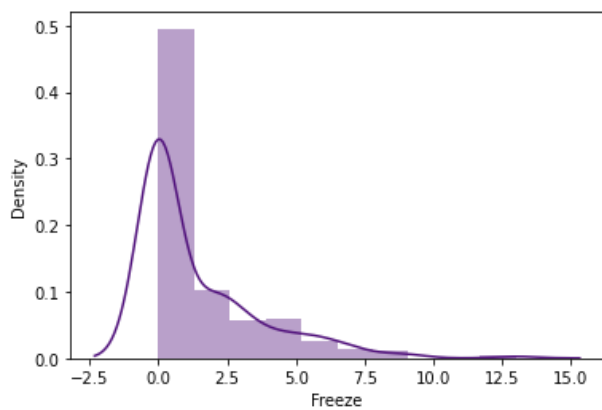


Image 5: Left Sided skew

Step 7: Using pairplot, get the below plot to show the relational pattern between the continuous variables.

Scatterplot shows the distribution pattern on the diagonal line. Eggs Matured Vs Embryos Transferred and Eggs Retrieved Vs Mature Eggs have linear relationship.

Step 8: Use countplot for Protocol to get the count of Protocol wise nos

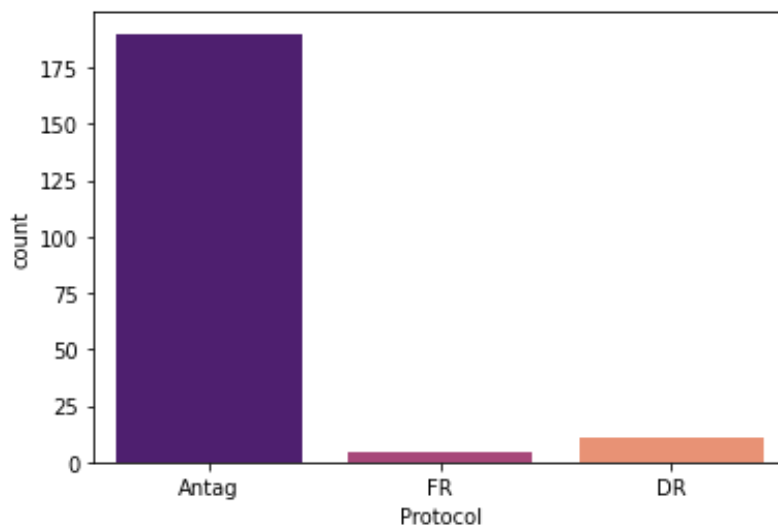


Image 8: Antagonist Protocol has been used maximum ~ 180 patients

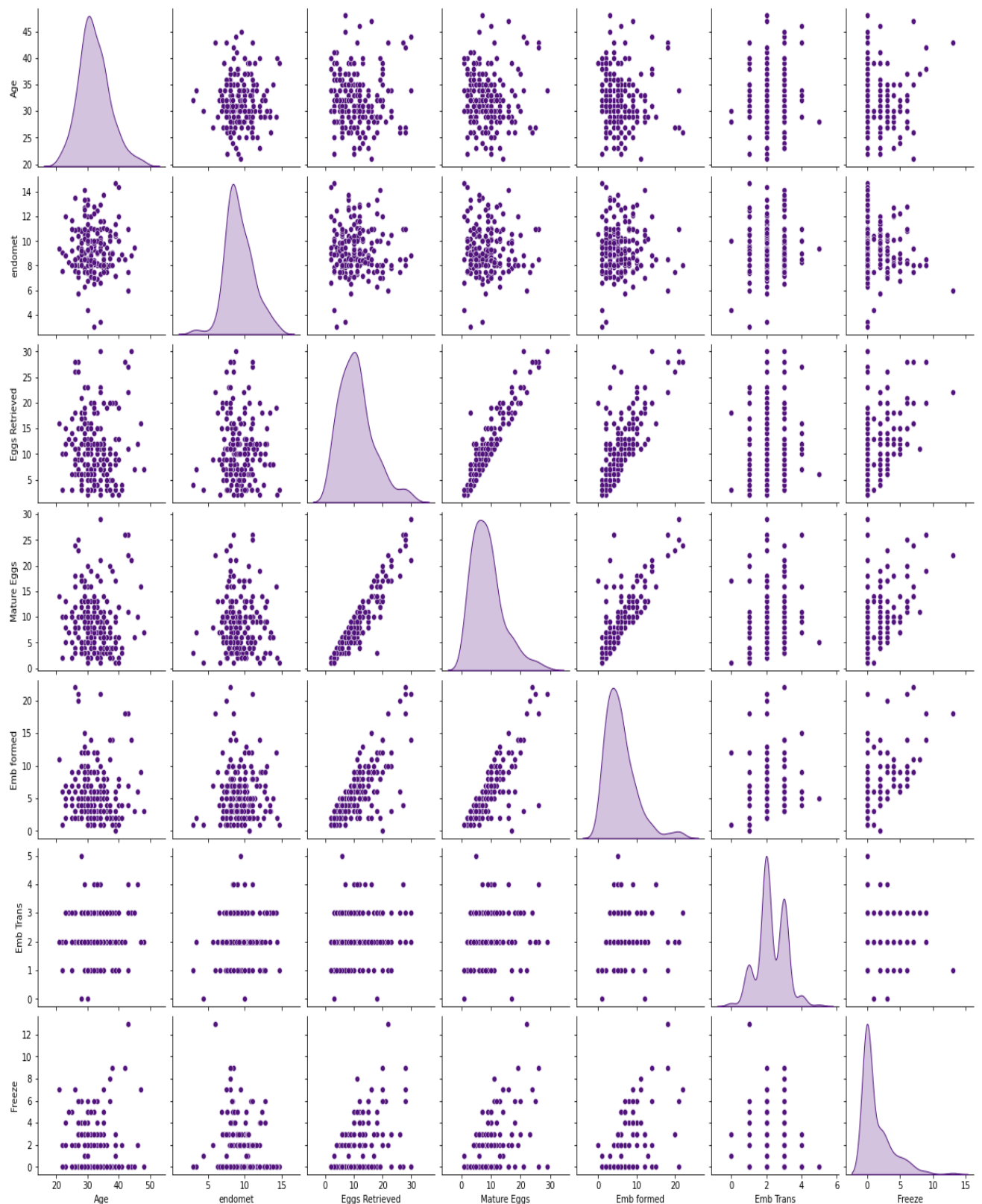


Image 7: Scatterplot showing bivariate relationship between the variables

Step 9: Make a boxplot to show the relation b/w Protocol used and the Eggs Retrieved

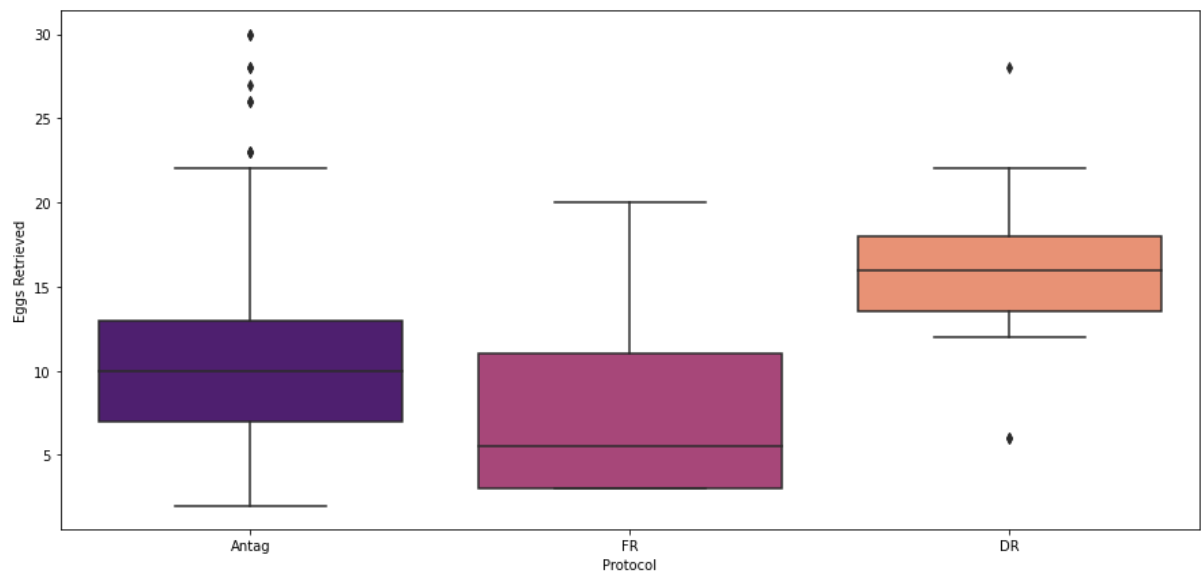


Image 9: The Boxplot shows median value for Eggs Retrieved is highest with Down Regulation and FR with lowest value.

Step10: Boxplot to show the relation b/w Embryos formed Vs Protocol used

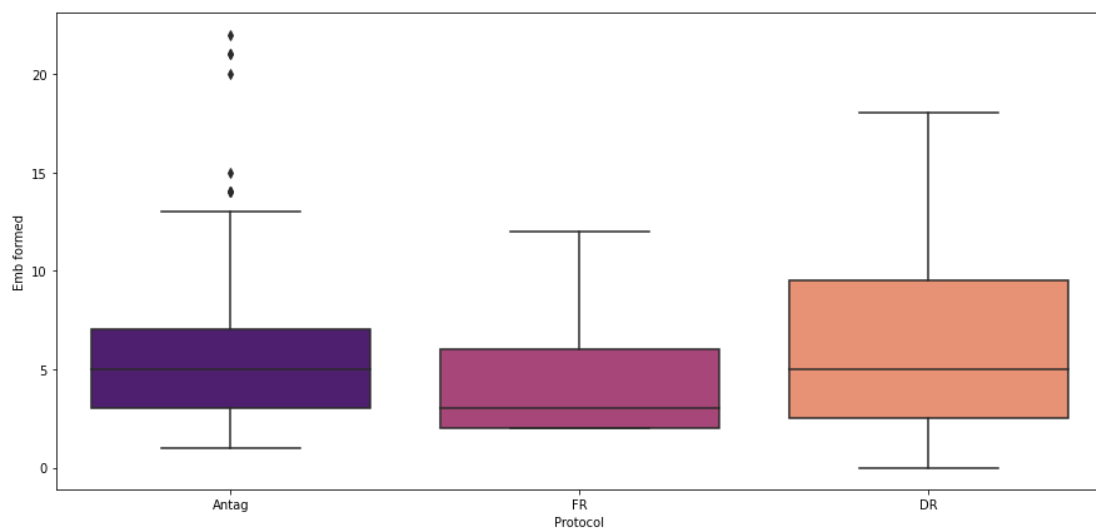


Image 10: Boxplot showing median values for embryos formed to be similar for DR and Antagonist Protocol

Step11: Plot the Pie Chart to show the %share for ET wise Pregnancy

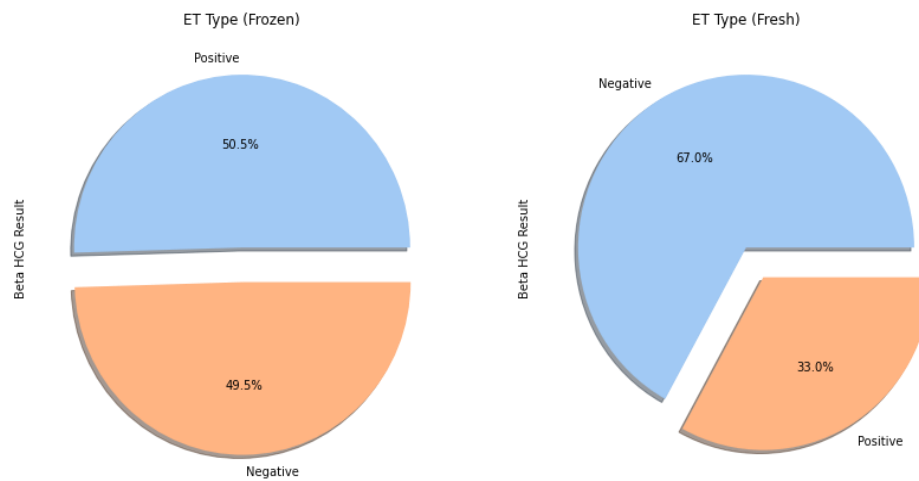


Image 11: Frozen ET has 50% Positive Pregnancy as compared to 33% Pregnancy rate with Fresh Transfers

Step12: Use crosstab to plot the barplot for Beta HCG Result, Emb formed and GnRH

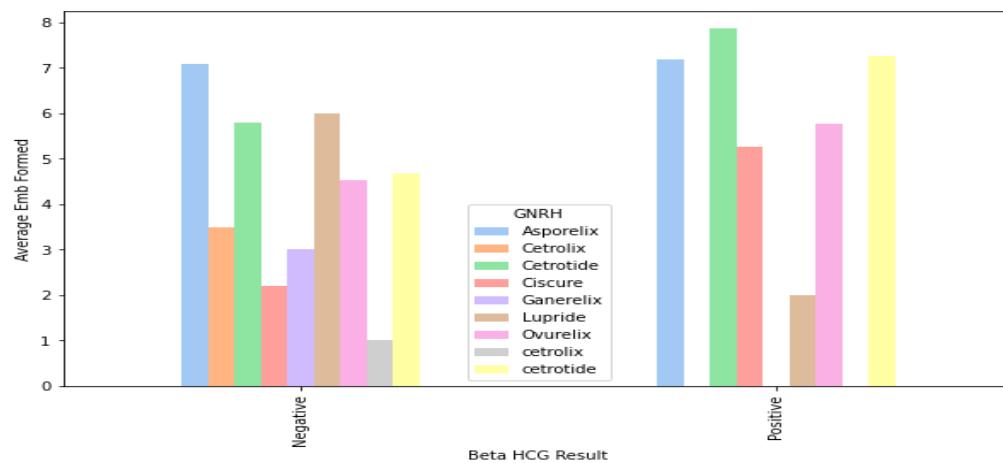


Image 12: Barplot shows higher positive fertility with the use of GnRH like Aspirelix and Cetrotide

Step 13: Barplot for ET Type, Beta HCG Result and Age

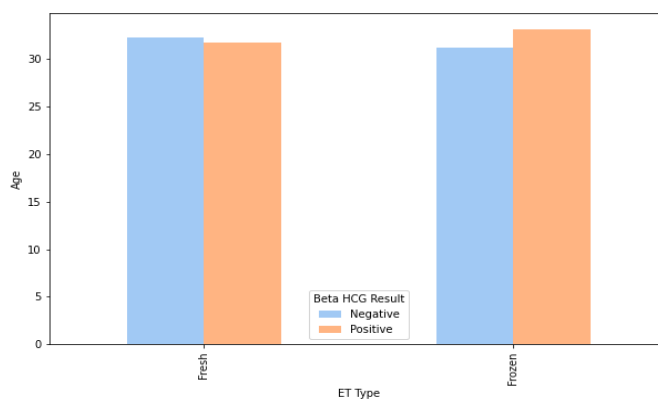


Image 13: Barplot shows Fresh ET has almost equal % of fertility rate with mean age at 32 Years. Frozen ET has higher positive pregnancy at mean age of 33 years

Step 14: Heatmap to show correlation between the variables

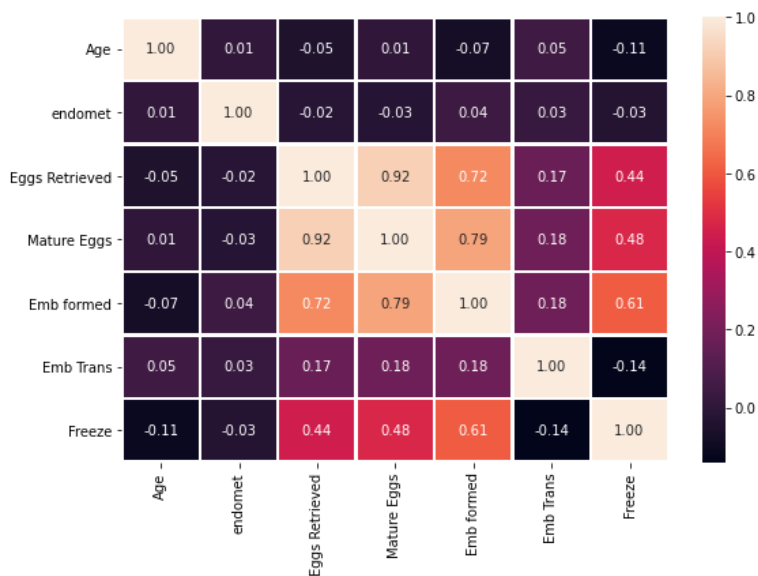


Image 14: Heatmap shows Multicollinearity is present between Eggs Retrieved and Embryos formed, mature eggs and embryos formed and embryos formed and freeze

Step15: Lm plot to show if linear relation is present between respective variables

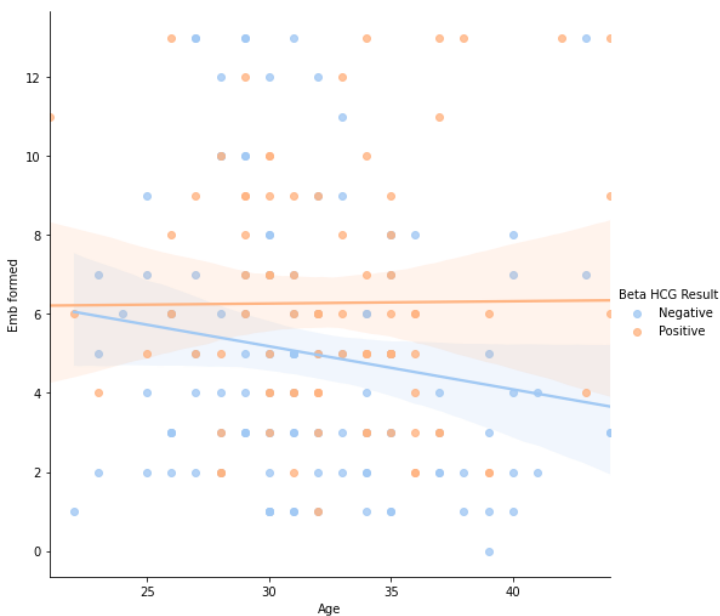


Image 15: Negative linear relationship is present between Age and Embryos formed with negative Beta HCG Result